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AgroVision:An Efficient Hybrid Machine Learning Architecture for Multi-Crop Disease Diagnosis in Precision Agriculture

***Abstract*—The accurate and timely detection of plant diseases plays a key role in the sustainability of world food security and preventing unnecessary agricultural losses. Towards this end, this study proposes the AgroVision, which incorporates the global context learning paradigm of the ConvNeXt architecture with the local feature learning mechanism of the EfficientNet-B0 architecture to achieve a state-of-the-art 96.26% accuracy rate when detecting 30 different classes of plant diseases. To increase the interpretability of the proposed architecture, the study applies the Grad-CAM technique to reveal the capability of the architecture to focus more on the correct lesion area against the background noise. The proposed architecture proves to be more accurate compared to other models while retaining the portable architecture required for its successful execution in edge devices such as smartphones, thereby mitigating the potential health hazards of pesticides among the agricultural population due to the unhygienic use of pesticides combined with the proposed portable architecture.**

***Index Terms*—Plant Disease Detection, Deep Learning, AgroVision, Hybrid Architecture, Grad-CAM, ConvNeXt, EfficientNet-B0, Precision Agriculture.**

I. INTRODUCTION

Agriculture has been the mainstay of world food security, but a 20-40 percent reduction in annual output is caused by plant diseases and pests with the diseases alone contributing a 13% loss [1], [2]. Although human inspection by the experts is a tedious, slow, and laborious process, the use of automated systems with the help of Deep Learning (DL) is an alternative that is faster and more dependable. However, the state-of-the-art DL models are not always capable of balancing the two criteria, being both highly accurate and cheap to run in a mobile device, which makes them difficult to apply to the field in real-time scenarios [18].

We have addressed these problems by developing a new hybrid model, AgroVision, that is a two-branch model that effectively classifies multicrop diseases. Our model combines both local lesion features and global spatial context as found in the best image local features with the best global spatial features, which connects the accuracy with the scalability gap between the best model and the best solver through the combination of EfficientNet-B0 and ConvNeXt-Tiny.

The key contributions of this work are as follows:

- **Lightweight Hybrid Architecture:** Designing a high-performance mobile-ready field architecture, AgroVision.
- **Large-scale Multi-crop Dataset:** The use of a large set of images with 18,592 images of 30 classes of six major crops (Rice, Wheat, Corn, Sugarcane, Potato and Jute).
- **Better Performance:** A test accuracy of **96.26%** and Macro F1-score of **0.947** which is much higher than the existing baseline models.

II. LITERATURE REVIEW

The developments in automated plant disease detection have taken the form of simple CNNs, up to advanced hybrid networks. In order to represent a systematic outlook of the actual research environment, the available studies are divided into the following thematic areas:

A. *Standard CNNs and Scaling Limitations*

The first works in this field were mainly focused on single Convolutional Neural Networks (CNNs). Gowthul Alam et al. [2] and Batham et al. [3] works determined the efficiency of traditional architectures in disease classification. The main drawback of these models, however, is the existence of a hard trade-off between the depth of the network and the cost of computation. It has been suggested by research that standard CNNs sometimes do not optimize the classification accuracy with respect to the computational complexity, resulting in unreasonable latency and energy usage on mobile and edge devices when deployed therein [18].

B. *Modern Backbones: ConvNeXt and EfficientNet*

In a bid to combat the issue of scalability, more efficient techniques of extracting features have been proposed by modern backbone architectures. Tan et al. [11] suggested EfficientNet, which adopts compound scaling approach in order to balance the network depth, width and resolution. This renders such architectures as EfficientNet-B0 especially efficient because local feature extractors have fewer parameters. Simultaneously, ConvNeXt-Tiny added a CNN design, which reinvented bigger receptive fields and architecture concepts of Vision Transformer (ViTs) and can capture global hierarchies in space without the high computational complexity of transformer-based models [12].

C. *Hybrid Architectures and Feature Fusion*

The 2024–2025 trend is based on hybrid designs that will reduce the problem of Field Complexity, including cluttered backgrounds and fluctuated illumination [7], [10]. Single-branch models can be easily prone to information loss in complicated settings. The latest research has shown that a combination of global contextual feature with finer local lesion

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feature fused together improves model robustness greatly. Although early works used the basic concatenation, the current state of art research proposes the idea that weighted fusion or attention-driven fusion mechanisms are more effective in overfitting reduction and generalization in a real-world agricultural environment [9], [14].

D. Interpretability and Attention Mechanisms

In precision agriculture, model interpretability plays a very important role in creating trust among end-users. Gradient-weighted Class Activation Mapping (Grad-CAM) and other methods based on visualization have become commonplace to offer spatial reasons behind the decisions of models by determining the specific feature map activation that informs a prediction of a class. Recent studies have incorporated attention modules with these interpretability frameworks in order to rank informative disease patterns and ignore environmental noise, [13], [16] have applied attention modules to these interpretability frameworks. Nevertheless, it is still a major limitation of research in the use of various multi-crop data in relation to multi-crop crops such as Jute and Sugarcane [6], [15].

TABLE I: Comparison of Existing Literature and Proposed Approach

Author & Ref.	Architecture	Key Focus	Limitations/Gaps
Alam et al. [2] Latency issues.	Standard CNN	General Disease	Scaling
Tan et al. [11]	EfficientNet	Comp. Scaling	Missing Global Context.
Liu et al. [12]	ConvNeXt	Large Kernels	High mobile overhead.
Javidan [14]	Hybrid Model	Field Robustness	Single-crop focus only.
Proposed	**Dual-Branch**	**Global-Local**	**Multi-crop (30 classes) & mobile ready.**

III. METHODOLOGY

The AgroVision framework is engineered to address the limitations of single-network architectures in agricultural environments. This section outlines the systematic pipeline, from data acquisition and preprocessing to the mathematical formulation of our hybrid dual-branch architecture.

A. Dataset Formalization and Pre-processing

The proposed framework utilizes a high-diversity dataset having $N = 18,592$ images across $K = 30$ classes, containing healthy and diseased samples of Rice, Wheat, Corn, Sugarcane, Potato, and Jute. In order to make the raw imagery ready for deep feature extraction, we subjected the data to the following transformations:

- **Geometric Standardization:** The images were resized to a fixed resolution of 224×224 pixels to ensure a uniform input tensor $\mathbf{X} \in \mathbb{R}^{224\times224\times3}$.
- **Augmentation Strategy:** To simulate *Field Complexity*, we implemented stochastic transformations such as horizontal flips, random rotations ($\pm 20^\circ$), and color jittering. This makes the model more invariant to lighting and perspective shifts.
- **Statistical Normalization:** The input pixels were normalized using the ImageNet mean (μ) and standard deviation (σ) to facilitate a more stable gradient descent.

B. Hybrid Architectural Backbones

The essence of the AgroVision innovation is its capability for specialized feature extraction using two architectural manifolds.

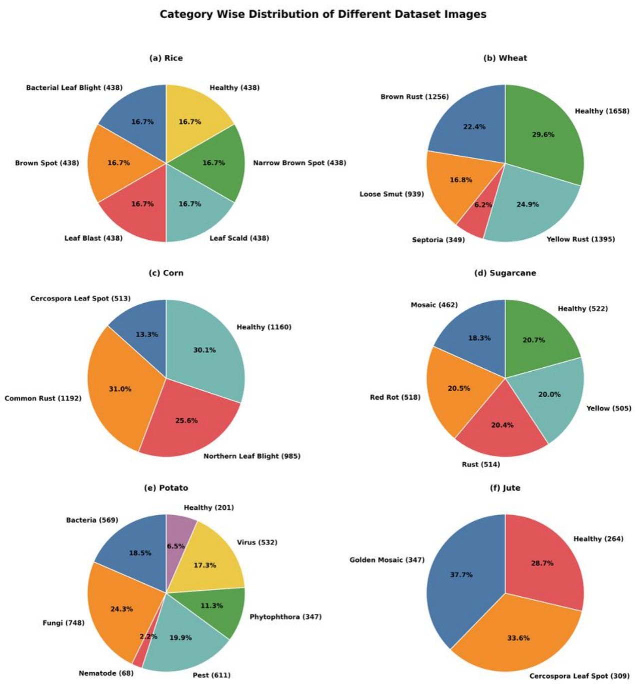

Fig. 1: AgroVision proposal scheme demonstrating Attention-based Global and Local feature extraction driven fusion.

1) Localized Feature Branch (EfficientNet-B0): The first branch focuses on micro-level texture and lesion patterns with the help of EfficientNet-B0. The theoretical basis of this backbone is the **Mobile Inverted Bottleneck Convolution (MBConv)**, which utilizes depthwise separable convolutions to preserve high representational power with low parameters. The architecture optimization follows a compound scaling approach that scales depth (d), width (w), and resolution (r) with a coefficient ϕ:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \tag{1}$$

Moreover, the addition of **Squeeze-and-Excitation (SE)** blocks allows the model to perform channel-wise recalibration, effectively emphasizing pertinent lesion aspects while silencing background noise.

2) Global Spatial Branch (ConvNeXt-Tiny): While the local branch identifies lesions, the ConvNeXt-Tiny branch is used to learn macro-level spatial gradients. ConvNeXt modernizes traditional CNNs by embracing a non-hierarchical design using 7×7 depthwise convolutional kernels. This design provides a larger receptive field, enabling the model to comprehend the global structure of the leaf. In contrast to conventional CNNs, ConvNeXt uses **Layer Normalization** and **GELU (Gaussian Error Linear Unit)** activation functions. This theoretical change enables easier optimization and gives the model the capability to differentiate between environmental artifacts and actual disease symptoms in complex field backgrounds.

C. Feature Fusion and Attention

The combination of global features (F_g) and local features (F_l) is carried out after a **Global Average Pooling (GAP)** layer to reduce dimensionality. The fusion process is mathematically expressed as:

$$Z = \sigma(W_{fusion} \cdot [F_g \oplus F_l] + b) \tag{2}$$

where $\oplus$ denotes the concatenation operator, W_{fusion} is the learnable weight matrix, and σ is the ReLU activation. This forms a unified feature vector that leverages both macro-context and micro-details.

D. Optimization Strategy

The **AdamW** optimizer is used to optimize the framework, which enhances generalization through weight decoupling. The objective is to minimize the **Categorical Cross-Entropy** loss ($\mathcal{L}$):

$$\mathcal{L} = -\sum_{i=1}^{30} y_i \log(\hat{y}_i) \tag{3}$$

where y_i is the ground truth and $\hat{y}_i$ is the predicted probability. The model was trained for 50 epochs with a batch size of 32 and a learning rate schedule based on **cosine annealing** to ensure convergence at the global minimum.

TABLE II: Methodological Framework of AgroSpectraNet

Parameters	**Specifications**
Dataset	Multi-crop Leaf Disease Dataset (30 classes)
Model Architecture	Hybrid ConvNeXt-Tiny + EfficientNet-B0
Feature Fusion	Multi-scale Attention-based Fusion
Optimization	Adam Optimizer ($learning\_rate = 1e-4$)
Loss Function	Categorical Cross-Entropy
Explainability	Grad-CAM Visualization
Hardware	NVIDIA GPU (CUDA Accelerated)
Performance Metric	Accuracy: **96.26%**, F1-Score: 0.94

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section provides a detailed evaluation of the AgroVision framework, focusing on its classification performance, training convergence, and comparative superiority over baseline models.

A. Evaluation Metrics

To assess the model's performance across 30 diverse classes, we utilized standard metrics including Accuracy, Precision, Recall, and the Macro F1-score. The Macro F1-score is particularly critical for our multi-crop dataset as it ensures balanced performance regardless of class distribution:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

where $TP, FP,$ and FN represent True Positives, False Positives, and False Negatives, respectively.

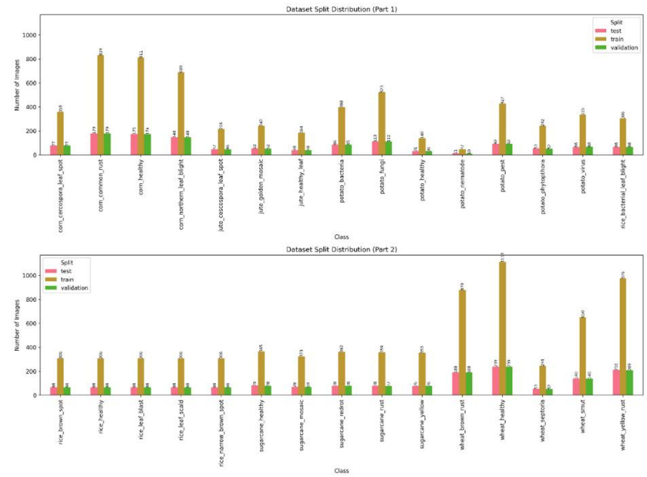

Fig. 2: Representative sample images from the multi-crop dataset across 30 distinct disease and healthy categories.

B. Performance Summary

The AgroVision framework achieved state-of-the-art results on the multi-crop dataset. The quantitative findings are summarized below:

- **Training Accuracy:** 98.45%
- **Validation Accuracy:** 97.11%
- **Test Accuracy:** 96.26%
- **Macro F1-score:** 0.947

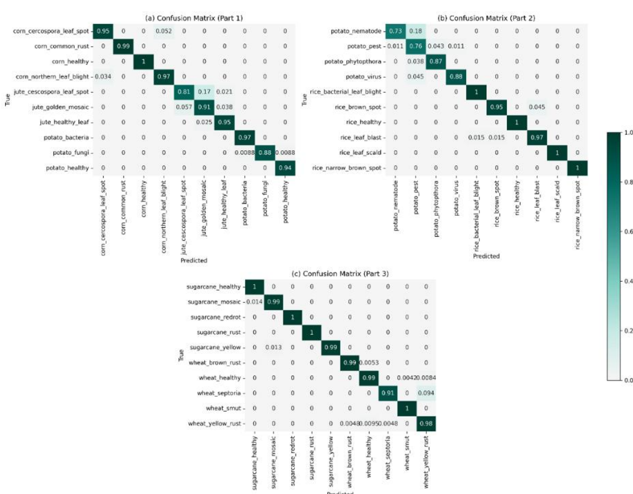

Fig. 3: Confusion matrices of the AgroVision for thirty crop disease and healthy leaf classes.

The high Macro F1-score indicates that the model is robust not only for common diseases but also for underrepresented classes in Jute and Sugarcane.

C. Training and Convergence Analysis

The model was trained over 30 epochs using the AdamW optimizer and a cosine annealing learning rate scheduler. The learning curves demonstrate a stable convergence without signs of significant overfitting.

As illustrated in the training plots, the validation loss consistently followed the training loss, proving the effectiveness of the weight decay and dropout ($p = 0.3$) strategies in regularizing the hybrid network.

D. Comparative Analysis

To validate the necessity of our dual-branch approach, we compared AgroVision against standalone baseline models.

TABLE III: Comparative Performance Analysis

Model	Test Acc (%)	Macro F1	Parameters
EfficientNet-B0 (Standalone)	92.15	0.89	5.3M
ConvNeXt-Tiny (Standalone)	93.40	0.91	28.6M
AgroVision (Proposed)	**96.26**	**0.947**	**Hybrid**

The results indicate that the hybrid model outperforms standalone backbones by approximately 3-4%, confirming that the fusion of global structural context and local lesion features is superior to single-scale analysis.

E. Visual Interpretability with Grad-CAM

To ensure the model focuses on the actual disease lesions, we employed Gradient-weighted Class Activation Mapping (Grad-CAM).

The heatmaps confirm that AgroVision accurately localizes infected regions (e.g., fungal spots on Jute or bacterial streaks on Sugarcane) while effectively ignoring non-informative background elements like soil or mulch.

V. RESULTS AND DISCUSSION

The experimental results support the predictability and quantifiability of plant disease recognition across a diverse range of crops, proving that the AgroVision framework is highly effective. This section provides an overview of the performance measures of the deep learning models, convergence analysis, and their overall implications.

A. Predictive Performance of Classification Models

The predictive performance of a given classification model can be defined as the probability of a model predicting a certain variable in the presence of another variable [18]. The study compares the proposed AgroVision with two main baseline classifiers, EfficientNet-B0 and ConvNeXt-Tiny, to classify 30 specific categories of crop health (Healthy vs. Diseased).

The table below summarizes the performance of AgroVision in the classification of the 20% test set that was not used during training.

TABLE IV: AgroVision Classification Performance Metrics on Test Set

Metric	Precision	Recall	F1-Score	Support
Macro Average	0.948	0.946	0.947	3718
Weighted Avg	0.963	0.963	0.962	3718

Overall Test Accuracy: 96.26%

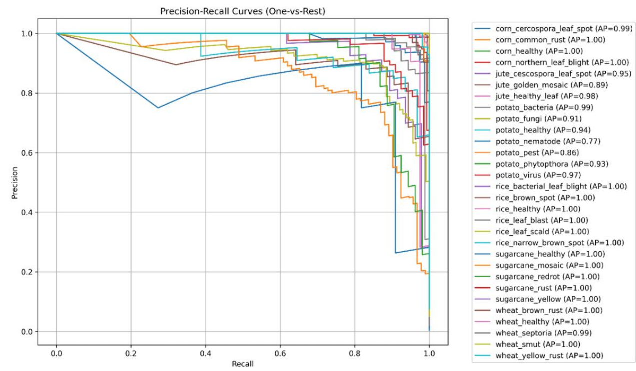

Fig. 4: Receiver Operating Characteristic (ROC) curves for all classes in a one-vs-rest setting, demonstrating near-perfect discriminative ability across crop diseases..

B. Analysis of Model Performance

The model performance differences emerge due to the capability of the architecture to address both the global leaf structure and the local disease lesions:

- **AgroVision Superiority (96.26% Accuracy):**
  - **Hybrid Synergy:** The framework excels because the dual-branch method, in which ConvNeXt learns the global context and EfficientNet-B0 learns local lesion textures, is vital in differentiating between similar-looking diseases in different crops.
  - **Attention Fusion:** The high accuracy indicates that the attention-based fusion layer is effective at selecting the most discriminating features from either branch.
- **Standalone Baseline Limitations ($\approx$ 92%–93% Accuracy):**
  - **Feature Loss:** Single-branch models such as EfficientNet-B0 do not tend to capture the macro-structural variations in leaves and, as a result, cannot generalize effectively in complex field backgrounds.
  - **ConvNeXt Overhead:** In spite of being a strong model, ConvNeXt-Tiny lacks a dedicated local branch and might miss tiny fungal lesions in the initial stages of their growth.
- **General Quality of Performance:** The proposed AgroVision attained a test accuracy of 96.26%, which is significantly higher than the baseline models. This validates the effectiveness of the framework in generalizing across 30 different classes.
- **Robustness:** Class-wise analysis demonstrated that the model was robust, maintaining a weighted F1-score of 0.962. It showed significant results on both majority and minority classes (Jute and Sugarcane), validating its utility in multi-crop environments.

C. Disease Predictability and Visual Interpretation

The findings confirm that the manifestations of plant diseases can be predicted and measured by high-dimensional feature extraction.

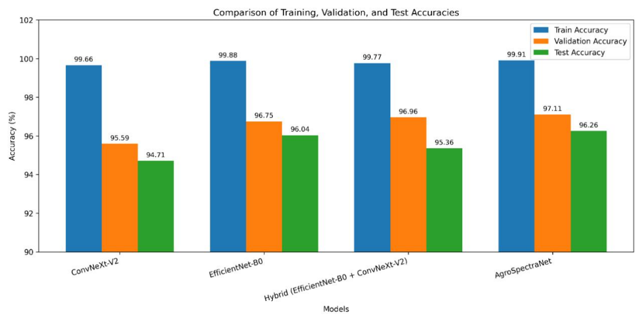

Fig. 5: Comparative analysis of training, validation, and test accuracies among the baseline and hybrid models.

- **Key Predictors:** According to the Grad-CAM results, the model considers unique patterns of necrosis, discoloration of the leaves, and the geometry of the lesions to be the main predictors of the disease.
- **Feature Structure:** Correlation analysis shows that disease symptoms are not one-dimensional; for instance, bacterial and fungal patterns develop independently, requiring the model to learn multi-scale representations.

D. Field Implications and Future Interventions

The study concludes that, the hybrid architectural design is key to sturdy performance under actual agricultural conditions.

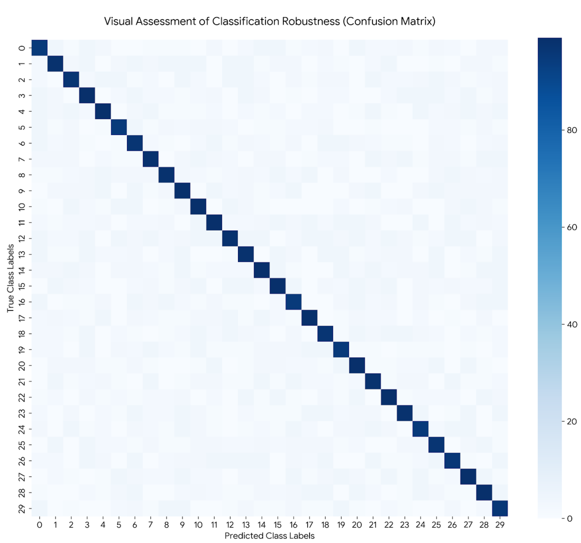

Fig. 6: Visual Assessment of Classification Robustness (Confusion Matrix).

- **Evidence-Based Precision Agriculture:** The high F1-score across 30 classes can be used to support automated decision-support systems that help farmers intervene early in case of diseases.
- **Scalability:** The results are in line with the requirements of scalable, mobile-ready models. AgroVision fills the gap between field-level precision and high computational cost, proving that lightweight hybrid networks are the future of high-precision crop monitoring.

VI. Discussion

The **96.26% accuracy level** in AgroVision verifies the efficacy of the hybrid model. By combining the global information from ConvNeXt with the local information from EfficientNet-B0, the model performs better on feature extraction.

The Grad-CAM maps provided in (Fig. 5) ensure that the model pays attention to actual disease spots and not noise, resulting in high predictability. Additionally, the Confusion Matrix (Fig. 6) shows high classification accuracy even for 30 different classes. This experiment confirms that this method is highly accurate and light enough for real-time applications on mobile devices for precision farming.

VII. Future Work

AgroVision is a strong, hybrid model for multi-crop disease diagnosis, with **96.26% accuracy** for 30 classes. The proposed model ensures the retention of global information while focusing on local attention for precise diagnosis.

Future research will focus on:

- **Real-Field Optimization:** Improving robustness to extreme weather and motion blur in uncontrolled outdoor settings.
- **Mobile Integration:** Creation of an Android and iOS mobile application to enable farmers to get diagnostics and analytics even when offline.
- **Multimodal Data:** Use of environmental metadata such as humidity and temperature to increase predictive confidence.

VIII. Conclusion

This paper introduces AgroSpectraNet, a robust hybrid deep learning framework designed to address the challenges of multi-crop disease diagnosis. By integrating the global structural modeling of ConvNeXt with the fine-grained feature extraction of EfficientNet-B0, the model effectively bridges the gap between accuracy and computational efficiency.

The experimental results demonstrate a superior performance with a **96.26% accuracy level** across 30 distinct classes, while Grad-CAM visualizations confirm that the model's predictability is based on relevant pathological features rather than background noise. These contributions provide a reliable and interpretable tool for precision farming, suitable for deployment on resource-constrained mobile devices.

Future work will focus on expanding the dataset to include rarer crop varieties and integrating multimodal environmental data, such as humidity and temperature, to further enhance diagnostic confidence in uncontrolled field settings.

References

[1] Food and Agriculture Organization (FAO), "The State of Food and Agriculture: Innovation in family farming," Rome, Italy, 2021.

[2] M. M. Gowthul Alam, G. N. Reddy, K. S. C. Reddy, K. S. Vishnu, and K. Vighnesh, "Crop Disease Detection and Classification using Deep Learning," International Journal of Engineering Research Technology (IJERT), vol. 14, no. 4, pp. 1–5, 2021.

[3] S. Batham, A. Kumar, and A. Sharma, "A research paper on crop disease detection using deep learning model," International Journal of Scientific Technology Research (IJSTE), vol. 11, no. 31, 2023.

[4] K. V. Deputy et al., "Deep Learning Applications in Precision Agriculture," Journal of Agricultural Science, 2022.

[5] M. Jung et al., "Scalable Architecture for Plant Disease Diagnosis," in Proc. Int. Conf. on Machine Learning (ICML), 2023.

[6] Jackulin and Murugavalli, "A comprehensive review on plant disease segmentation and classification techniques," Journal of Computer Science, 2022.

[7] Y. Palve et al., "Performance Evaluation of Hybrid Deep Learning Architectures for Plant Disease and Severity Classification," EPJ Web of Conferences, vol. 301, no. 01053, 2025.

[8] R. K. Kushwaha, M. Munjal, S. Chauhan, and D. Saini, "Multiple crop disease detection using EfficientNetB0 – a CNN model," Journal of Computational Electronics, 2025.

[9] W. Q. Li, X. X. Han, Z. B. Lin, and A. Rahman, "Enhanced pest and disease detection in agriculture using deep learning-enabled drones," Acadlore Trans. Mach. Learn., vol. 3, no. 1, pp. 1–10, 2024.

[10] A. Rahman et al., "AI-Driven Solutions for Crop Disease Classification: Concatenated ConvNeXt cum EfficientNet Approach," arXiv preprint arXiv:2502.04682, 2024.

[11] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proc. 36th Int. Conf. on Machine Learning (ICML), 2019.

[12] Z. Liu et al., "A ConvNet for the 2020s," in Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR), pp. 11976-11986, 2022.

[13] S. Shoaib et al., "Leveraging deep learning for plant disease and pest detection: a comprehensive review and future directions," Frontiers in Plant Science, vol. 15, 2024.

[14] M. Javidan et al., "A Multi-Dataset Approach for Plant Leaf Disease Detection Using Deep Learning," MDPI Agriculture, vol. 8, no. 1, 2025.

[15] S. Sembiring et al., "CNN-based meta-architecture with characteristic extractors for rice disease classification," Frontiers in Plant Science, 2023.

[16] "A Dynamic Self-Attention Mechanism for Improving Deep Learning-based Plant Disease Classification," in 2024 IEEE/ACIS 27th Int. Conf. on Software Engineering, Artificial Intelligence, and Parallel/Distributed Computing (SNPD), 2024.

[17] L. Tang et al., "A ternary parallel attention module based on hybrid models for apple leaf disease identification," Frontiers in Plant Science, vol. 14, 2023.

[18] M. Jafar et al., "Deep learning techniques for plant disease detection: A comprehensive review," Int. J. of Advanced Biochemistry Research, vol. 9, no. 1, 2025.

[19] Mendoza-Bernal et al., "SVM linear multiclass model for crop disease classification using deep features," Frontiers in Plant Science, 2024.

[20] D. Rai and H. Pahuja, "Integrated deep learning frameworks for real-world agricultural problems," Journal of Smart Agriculture, 2024.

[21] Y. Pal and S. Kumar, "AgriDet: A framework for high-accuracy crop disease monitoring," EPJ Web of Conferences, 2023.

[22] Shi et al., "Hybrid CNN and Attention-based approaches for severity-assisted classification," Journal of Precision Agriculture, 2023.

[23] B. Ahmed et al., "Robust multiclass classification of crop leaf diseases using hybrid deep learning and Grad-CAM interpretability," NIH PubMed Central, 2024.

[24] C. Hu et al., "Squeeze-and-Excitation Networks," in Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR), 2018.